Parvathareddy Sai Manogna

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.61
Bashyam Jr. College, Guntur	Class XII (BIEAP)	2024	98.80%
Sri Chaitanya High School, Bapatla	Class X (BSEAP)	2022	98.00%

PROJECTS

Low-Latency Limit Order Book Matching Engine

Personal Project - C++, Standard Template Library (STL), Algorithmic Optimization

- Core Architecture: Developed a fast matching engine to execute buy and sell orders based on strict price-time priority, accurately handling partial fills and dynamic market spreads.
- Algorithmic Optimization: Engineered constant-time order cancellations (O(1)) by combining hash maps with doubly-linked lists, eliminating the need for slow linear searches.
- Data Structure Design: Utilized self-sorting tree structures to automatically manage price levels, guaranteeing instant access to the highest bids and lowest asks.

Enhanced xv6 Operating System

Course project - *C, RISC-V, QEMU, Linux*

- Extended the xv6 RISC-V kernel by designing and integrating a 4-level SC-MLFQ CPU scheduler.
- Built a virtual memory subsystem featuring lazy allocation and a Clock page replacement algorithm.
- Upgraded the storage layer to support disk-backed swapping, FCFS/SSTF disk scheduling, and RAID storage.
- Implemented custom system calls to enable deep process accounting and real-time memory tracking.

Finite Difference Solver for Partial Differential Equations

Course Project - Computer Aided Numerical Methods | *C, Python, NumPy, Matplotlib*

- Implemented finite difference solvers in C for 1D/2D advection, diffusion, and Burgers' equations, using explicit, implicit, and ADI-based numerical schemes.
- Performed stability and convergence analysis across varying Péclet regimes and visualized numerical solutions using Python (NumPy, Matplotlib)

SKILLS

Programming Languages : C, C++, Python, SQL

Tools : Git, GitHub, Linux, LaTeX

Libraries : NumPy, Matplotlib

Core Concepts : OOP, DSA, Operating Systems, DBMS

RELEVANT COURSES

Statistics & Mathematics: Probability, Introduction to Statistics, Linear Algebra, Calculus, Differential Equations, Numerical Methods, Transform Techniques, Complex Variables

Computational & Computer Science: Fundamentals of Scientific Computing, Computer Aided Numerical Methods, Data Structures and its Applications, Introduction to Programming, Operating Systems

Other: Sustainability and Climate Change, Computational Methods in Material Science, Electronic Structure and Atomistic Modeling of Materials

SCHOLASTIC ACHIEVEMENTS

- Secured A in Introduction to Probability
- Secured A in Differential Equations
- Secured A- in Operating Systems

POSITIONS OF RESPONSIBILITY

Events Head (Previously Events Coordinator), Elan & nVision - IIT Hyderabad's Annual Techno-Cultural Fest

- Promoted from Events Coordinator to Events Head based on performance and leadership.
- Leading a team of 10+ coordinators to drive end-to-end execution of flagship events.
- Overseeing event strategy, logistics, budgeting, and cross-functional coordination.
- Streamlining planning and task allocation, improving coordination across teams and ensuring timely execution of flagship events.